27 — Transit + Anchor

Proves · Transit plus Anchor — a second mode with hotel load but no propulsion.

Mechanics this scenario turns on

- Every active mode runs its **own** Level 1 — own demand, own combinations, own baseline, own optimum, own t/h. There is no single "tonnes per hour" for the vessel; the year is `Σ (mode t/h × mode hours)`.
- Level 2 and Level 3 are computed for **Transit only** (decision D4/Q5 — no workbook counterpart elsewhere). Other modes get an empty result, and the pinned-baseline radio does not reach them.
- With several modes the Power Demands tables gain one row per mode, and the header is an **hours-weighted average** (total energy ÷ total hours), not a sum.
- Savings are a **difference**, so a mode whose baseline equals its optimum contributes zero — it is present on both sides and cancels, not excluded.

Panels described below · What Anchor is, in the model · The plant and the numbers · The SG-forced rule is visible here

Anything not described here — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 07.

Anchor mode had never been exercised on its own — it appeared only inside scenarios 11 and 14, bundled with three other modes, where its contribution cannot be isolated.

What Anchor is, in the model

```
propulsion : 0        (the vessel is not moving)
hotel      : as entered
sea margin : not applied — only Transit adds it
```

Anchor and Port are the two zero-propulsion modes. The difference between them is only the hotel figure the user enters; the code path is identical.

The plant and the numbers

Scenario 03's vessel plus **1 500 h at anchor with a 600 kW hotel load**.

```
Transit   5 000 h   propulsion 12 036.2   SG 3 250.0   AE 626.0
Anchor    1 500 h   propulsion      0.0   SG   600.0   AE   0.0
```

Baseline **13 810.2 t/yr** · L1 savings **192.72 t/yr** · L2 and L3 zero (one aux engine).

The SG-forced rule is visible here

At anchor the shaft generator carries the whole 600 kW hotel load — which means **a main engine is running to spin it**, at roughly 2 % load, while four idle auxiliary engines sit unused.

That is logged observation #1: when a shaft generator is installed, `Level1CandidateBuilder` requires every combination to run it, and an SG cannot turn without its main engine. Real vessels at anchor run an auxiliary generator instead.

The anchor hours therefore inflate the annual fuel figure, and the more of them a vessel has the worse the distortion. Scenario 27 exists partly to make that measurable: compare its baseline with scenario 03's (13 652.6 → 13 810.2 for 1 500 anchor hours) and the cost of the rule is right there.

Takeaway: Anchor is a simple mode with a complicated side effect. Isolating it turns a known observation into a number.
